

PRINCIPLES OF SURGICAL ONCOLOGY

CHAPTER

1

Benign Breast Diseases

Benign breast diseases represent the majority of breast lesions (88%). The incidence starts and increases in the second decade of life, then decreases during the fourth and fifth decades, opposite to the malignant lesions, which continue to increase even after menopause. These lesions never turn malignant

ANATOMY OF THE BREAST

Composition of the Breast

- The *mammary gland* is a modified apocrine gland situated on the thoracic cage from the 2nd to the 6th rib. It is present in the superficial fascia with no covering capsule
- The *areola* is the rounded area surrounding the nipple it is faint in color and then becomes dark. It contains Montgomery follicles that secrete sebaceous material to lubricate the nipple during lactation
- The *nipple* is an elevated part in the center of the areola. It is situated at the level of the 4th intercostal space just outside the mid-clavicular line pointing forward and outward. It contains the orifices of the lactiferous ducts.
- The *gland* is composed of 16-20 lobes; each lobe contains adipose tissue and lobules, each of which is formed by acini of glandular tissue. Each lobe has its own lactiferous duct that opens by a separate orifice in the nipple with a dilatation named “lactiferous sinus”. The gland is supported by fibrous strands; Cooper’s ligaments that support the gland and divide it into lobes
- Three muscles are distinguished behind the breast: pectoralis major, pectoralis minor and serratus anterior

Vascularity of the Breast

- Axillary artery gives lateral thoracic artery and associated with thoraco-acromial artery and lateral thoracic artery
- Internal mammary artery: from subclavian artery and gives perforators behind the first 6 ribs
- Posterior intercostal artery: from the aorta supply from 6-9 intercostal spaces
- Lymphatic drainage: Axillary lymph nodes (LNs) drain 80% of the breast, while the internal mammary LNs drain the remaining 20%

CONGENITAL ANOMALIES

- *Athelia*: absence of the nipple
- *Polythelia*: accessory nipples (multiple nipples)
- *Retraction*: inversion of the nipple
- *Amastia*: absence of the breast and of the nipple. It should be differentiated from acquired retraction due to chronic breast abscess, duct ectasia and breast cancer.
- *Amazia*: absence of the breast with presence of the nipple
- *Hypoplasia*: under-development of the breast
- *Mammary hypertrophy* and *ectopic breast* (accessory breast)
- *Polymastia*: super-numerary breast along the milk line

NIPPLE DISCHARGE

- *Clear*: fibrocystic disease (FCD) or cancer
- *Bloody*: duct papilloma or cancer
- *Green or black*: duct ectasia or fibroadenosis
- *Pus (creamy)*: abscess
- *Yellowish*: colostrum
- *Milky*: lactation or hyper-prolactinemia

BREAST INFECTION

- Neonatal mastitis
- Pubertal mastitis
- Acute mastitis and breast abscess
- Chronic breast abscess
- Tuberculous mastitis

Neonatal Mastitis

- During the first or second week after birth
- Staph aureus is usually the causative organism
- Characterized by redness, erythema, tenderness with breast enlargement sometimes up to abscess formation associated with high fever
- *Treatment*
 1. Stage of cellulitis: conservative
 2. Stage of abscess formation: drainage

Acute Mastitis and Breast Abscess

- Infection that occurs during lactation or late months of pregnancy
- Caused by staph aureus
- May be preceded by a fissure in the nipple through which the organism is introduced

Stage of Engorgement

- Milk is retained in the main ducts inviting the bacterial ascending infection

Stage of Mastitis

- *Clinical presentation:* fever + engorged enlarged breast with erythema, redness, tenderness and skin edema. About 30% will develop abscess
- *Treatment*
 1. Prevention: good hygiene of the nipple during lactation
 2. Evacuation of the engorged breast
 3. Antibiotics and analgesics

Stage of Abscess Formation

- Throbbing pain
- Local signs of inflammation
- Skin edema
- Should not wait for fluctuation
- Axillary lymph node may be tender
- *Drainage of the abscess*
 - o Under general anesthesia
 - o Through circumareolar or infra-mammary incision
 - o By Hilton method of drainage
 - o Stop feeding from the infected breast until inflammation subsides
 - o Lactation from the sound breast
 - o Manual evacuation of the breast or by pump
 - o Antibiotics according to culture and sensitivity

Chronic Breast Abscess

- Painful breast mass with fibrosis and thick skin
- It may sometimes simulate malignant mass and hence should be investigated by mammogram and excisional biopsy if needed
- It may be due to:
 1. Inadequate drainage of acute abscess
 2. Inadequate intake of antibiotic after acute abscess
 3. Acute abscess left to be spontaneously drained without adequate medication

Tuberculous Mastitis

- May spread through axillary lymph nodes or directly through infected ribs or parasternal junctions or pleural Common in low socioeconomic communities
- *Clinical presentation*
 1. 50% axillary sinus
 2. Acute abscess due to secondary infection
 3. Multiple discharging breast sinuses and scars
 4. Systemic manifestations of TB
 5. Approximately, 25% of cases are associated with pulmonary TB

- *Management*
 1. Proper investigations looking for TB organism
 2. Anti-tuberculous therapy
 3. Excisional biopsy of a localized mass under covered by anti-TB drugs

FIBROCYSTIC DISEASE (FCD)

- Synonyms: ‘fibroadenosis’, ‘ANDI’ (aberration of normal development and involution)
- It is a benign breast disease that affects pre-menopausal females and occurs as a result of cyclic and estrogen hormone changes that occur in the premenopausal fertile age group
- It is manifested by (1) adenosis, (2) epithelial proliferation, (3) fibrosis, and (4) cyst formation

Clinical Criteria of Fibroadenosis

- Mastalgia: usually cyclic
- Discharge per nipple : any color but not bloody
- Unilateral or bilateral diffuse or sectoral cystic swelling

Management

- Mammogram and US
- Reassurance: sufficient in 85% of cases
- Anti-inflammatory drugs
- Bromocriptine (anti-prolactin)
- Danazol (anti-gonadotrophin)
- Aspiration of the cyst and cytology
- Excisional biopsy of a localized mass $\pm$ frozen-section biopsy

FIBROADENOMA

Etiology

- Considered as aberration changes that occur in the mammary gland after the age of puberty
- It is derived from the terminal duct lobular unit (TDLU) as a result of estrogen instability during the fertile age group

Pathology

- Well-defined due to the presence of well-formed capsule
- Its size varies from 1 cm up to 5-20 cm (*giant fibroadenoma*)
- May be solitary or multiple
- Microscopically formed of a mixture of ductal and fibrous tissue
- Soft stroma, large compressing on the ducts (*intra-canalicular*) or hard, rounded or oval (mouse of the breast) (*peri-canalicular*)

Diagnosis

- It is always diagnosed clinically Only doubtful lesions may need US and FNAC

Treatment

- *Reassurance* (never turns malignant)
- *Surgical excision* through a circumareolar, or if huge, infra-mammary incision

Phylloides Tumor

- It is a kind of *giant fibroadenoma* that affects females above 35 years of age
- It has the potential to be malignant
- Represents 1-2% of all breast lesions

Giant Fibroadenoma	Phylloides Tumor
- Stroma Hypo-cellular - Homogenous cellular configuration - No mitotic activity	- Stroma Hyper-cellular - Cellular Pleomorphism - Mitotic activity

Microscopically

- *Benign*: if < 4 mitosis per high power field
- *Malignant*: if > 4 mitosis per high power field or there is marginal infiltration and necrosis. Only 15% of all phylloides are malignant; 20% of which gives metastasis mainly to the lung

Diagnosis: FNAC or Tru-cut biopsy

Treatment: Excision biopsy for benign type and simple mastectomy for malignant and recurrent types

DUCT ECTASIA

- It is a benign breast condition that resembles malignancy if manifested by nipple retraction

Pathophysiology

- Blockage of the main lactiferous ducts by debris leading to chronic ductal dilatation, greenish discharge, stasis and inflammation (plasma cell mastitis)

Clinical Presentation / Diagnosis

- Pre-menopausal females
- Age < 40 years
- Smokers (even passive smokers)
- Non-bloody nipple discharge.
 - o Spontaneous or by expression
 - o Unilateral sometimes bilateral
 - o Yellow or green or brown or creamy

- Peri-areolar tenderness, sometimes associated with signs of inflammation, may form a peri-areolar or retro-areolar abscess
- Nipple retraction (slit like) due to chronicity of the condition, abscess formation and recurrence
- Mammogram and US: reveal retro-areolar main duct dilatation

Treatment

- *Conservative:* Stop smoking, anti-inflammatory drugs, drain small abscesses
- *Surgical:* Major duct excision by a circumareolar incision

TRAUMATIC FAT NECROSIS (TFN)

- *Etiology:* Rare condition that occurs in large fatty breasts due to chronic minor trauma (50% of cases)
- *Clinical presentation:* Hard, may be, tender ill-defined mass
- *Differential diagnosis:* breast cancer
- *Diagnosis:* triple assessment
 1. Mammogram
 2. US
 3. FNAC or Tru-cut
- *Treatment:* Excisional biopsy

DUCT PAPILLOMA

- Benign condition (never turns malignant)
- Presents in main lactiferous ducts (80%)
- Unilateral, uni-ductal, spontaneous, bloody nipple discharge
- May be associated with a retro-areolar mass
- *Diagnosis:* Mammogram and US. If there is a mass, FNAC or Tru-cut should be done
- *Treatment:* major duct excision

GALACTOCELE

- Large cystic breast lesion
- Affects females that suddenly stop lactation
- The skin overlying is stretched but normal and not attached
- Diagnosed clinically
- US confirms the condition
- Aspiration of milk may cure the condition
- Excision of the cyst (may be complicated by milk fistula)

CHAPTER

2

Breast Cancer

Incidence

- The incidence of breast cancer varies greatly around the world, being lower in less-developed countries and greatest in the *more-developed* countries.
- The general impression is that its incidence is increasing; it is common in women over 30 years; the incidence increases by increase in age.
- It is the 2nd leading cause of cancer-related death in females
- It comprises about 35% of all women cancers in Egypt
- Breast cancer was expected to cause 40,910 deaths in the US (7% of cancer deaths; almost 2% of all deaths)

Etiology

- No etiology is known for 95% of breast cancer cases,
- While approximately 5% of new breast cancers are attributable to hereditary syndromes.

Epidemiology and Risk Factors

- The definite cause is not known; however, there are certain risk factors.
- Race: Developed countries > developing countries, white > Black
- Age: > above 40 years
- Family history: 5 %
- Menstrual history: early menarche and late menopause
- Marital status: single > married
- Pregnancy: nulliparous > multiparous
- Past history of uterine carcinoma
- Mammary dysplasia: with proliferative changes are good candidates for screening

Clinical Features

Symptoms

1. Painless lump, accidentally discovered (commonest presentation)
2. Others:
 - Nipple discharge or retraction. Paget's disease
 - Skin changes (peau d'orange, dimpling, fungation, ulcers and nodules)

- Cancer en cuirasse
- Inflammatory breast cancer
- Axillary lymph nodes (LNs) (occult) or supra-clavicular mass
- Mastalgia (breast pain)
- Arm edema
- Low back pain, bone swelling (bone metastases)
- Cough or hemoptysis
- Krukenberg tumor and Plummer's shelf

Signs

1. A hard mass with ill-defined edges, not tender, may be fixed.
2. Lymph nodes are hard, not tender, early mobile, later on fixed.

Differential Diagnosis

- The diagnosis of late cases is easy, but the most important is the diagnosis of early cases. The following diseases may be missed as cancer:
 1. Fibrocystic disease (FCD).
 2. Mammary duct ectasia (MDE).
 3. Chronic breast abscess.
 4. Traumatic fat necrosis (TFN).

Pathology

- Adenocarcinoma of duct cell origin (TDLU) infiltrates the breast parenchyma.
- Left > Right (60:40)

Pathological Types

1. *Carcinoma of the nipple (Paget's disease) (1%)*
 - An eczematous lesion of the nipple and areola, most probably ductal carcinoma.
 - It grows outwards as well as inwards.
 - It carries relatively good prognosis because of early discovery.
2. *Carcinoma of the ducts (89%)*
 - Non-infiltrating (ductal carcinoma in-situ - DCIS) (1%):
 - a) Papillary type
 - b) Comedo Type.
 - Infiltrating duct carcinoma (IDC) (88 %):
 - a) Papillary type (1%)
 - b) Comedo type (1%)
 - c) Adenocarcinoma with fibrosis (78%)
 - d) Medullary cancer (4%)
3. *Lobular carcinoma (8%)*
 - It arises in the mammary lobules from the cells of the acini and the terminal ducts.

Spread

- Early breast cancer was considered a disease which spreads first to the regional lymph nodes (LNs), then to the blood stream. Now breast cancer is a systemic disease at the moment it presents. Cancer cells (micro- metastases) are present in the circulation from the early beginning. Metastases start to appear when there is weakness in the defence mechanisms of the patient.
- Routes of spread out:
 1. Local spread
 2. Lymphatic spread (by embolization and permeation)
 3. Blood (bones, liver, lung, brain , etc)

TNM Staging (UICC)

Primary Tumor (T)

Tx	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma in situ: intra-ductal carcinoma, lobular carcinoma in-situ, or paget's disease of the nipple with no tumor
T1	Tumor 2 cm or less in greatest dimension
T1a	0.5 cm or less in greatest dimension
T1b	> 0.5 cm but not > 1 cm in greatest dimension
T1c	> 1cm but not > 2cm in greatest dimension
T2	Tumor > 2 cm but not > 5 cm in greatest dimension
T3	Tumor > 5 cm in greatest dimension
T4	Tumor of any size with direct extension to chest wall or skin:
T4a	Extension to chest wall
T4b	Edema (including peau d'orange) or ulceration of the skin of the breast or satellite skin nodules confined to the same breast
T4c	Both (T4a and T4b)
T4d	Inflammatory carcinoma

Regional Lymph Nodes (N)

Nx	Regional LNs cannot be assessed (e.g. previously removed)
N0	No regional LN metastasis
N1	Metastasis to movable ipsilateral axillary LNs
N2	Metastasis to ipsilateral axillary LNs fixed to one another or other structures
N3	Metastasis to ipsilateral internal mammary LNs

Distant Metastasis (M)

Mx	Presence of distant metastasis cannot be assessed
M0	No distant metastasis
M1	Distant metastasis (includes metastasis to ipsilateral supra-clavicular LNs). The chest wall includes ribs, intercostal muscles, serratus anterior, but <i>not</i> pectoralis major

Stage Grouping

Stage 0	Tis	N0	M0
Stage 1	T1	N0	M0
Stage IIA	T0	N1	M0
	T1	N1	M0
	T2	N0	M0
Stage IIB	T2	N1	M0
	T3	N0	M0
Stage IIIA	T0	N2	M0
	T1	N2	M0
	T2	N2	M0
	T3	N1	M0
	T3	N2	M0
Stage IIIB	T4	Any N	M0
	Any T	N3	M0
Stage IV	Any T	Any N	M1

Breast Cancer Screening

- Breast cancer screening is an attempt to find unsuspected cancers.
- The most common screening methods are:
 1. Breast self-examination (BSE)
 2. Clinical breast examination
 3. X-ray mammography and ultrasonography
 4. Magnetic resonance imaging (MRI).
 5. Genetic testing may also be used.
- About 8 in 10 lumps discovered by BSE are harmless. Nevertheless, any abnormality thus detected should promptly be reported to a doctor.
- Women can easily miss a breast lump that an expert can find. For the same reasons it is better to learn BSE from an expert.

Mammogram

- Mammography is still the modality of choice for screening women for early detection
- Approximately, 85%-90% of all breast cancers are detectable by mammography. Early detection by mammography has reduced the mortality rate from breast cancer by 20%-30% in women over 50 years of age.
- Routine mammography of women over 40 years (USA) or 50 years (UK) is recommended and over 35 years if at risk.
- Computer-aided diagnosis (CAD) Systems may help in detecting breast cancer in an early stage. CAD is especially established in USA and the Netherlands.
- Digital mammogram
- Mammogram with contrast

Ultrasound

- Ultrasound is typically used for further evaluation of masses found on mammography or palpable masses not seen on mammograms.

MRI

- Proposed indications for using MRI for screening include:
 1. Strong family history of breast cancer
 2. Patients with BRCA-1 or BRCA-2 oncogene mutations
 3. Evaluation of women with breast implants
 4. History of previous lumpectomy or breast biopsy surgeries
 5. Axillary metastasis with an unknown primary tumor
 6. Very dense or scarred breast tissue

Diagnosis

Triple Assessment

1. History and clinical examination.
2. Mammogram and ultrasonography.
3. Fine needle aspiration cytology (FNAC), Tru-Cut biopsy, and excisional biopsy.

Biological Types of Breast Cancer

	Luminal A	Luminal B	Her-2/neu	Basal-like
Gene expression pattern	Expression(LMW) cytokeratins, and high expression of HR's and associated genes	Expression (LMW) cytokeratins, and moderate to weak expression of HR's and associated genes.	High expression of Her-2/neu . Low expression of ER and associated genes.	High expression of basal epithelial genes, basal cytokeratins. Low expression of ER and Her-2/neu associated genes.
Clinical	~ 50% of invasive breast cancer	~ 20% of invasive breast cancers	~15% of invasive breast cancers	~15% of invasive breast cancers
ER/PR status	ER/PR positive	ER/PR positive	ER/PR negative	Most ER/PR negative
Her-2/neu status	Her-2/neu negative	Her-2/neu expression variable (+/-)	Her-2/neu positive (by definition)	Her-2/neu negative("triple negative")
Biological features		High proliferation than luminal A	High proliferation	High proliferation

Treatment

- 1- Surgery
 - 2- Radiotherapy,
 - 3- Chemotherapy,
 - 4- Hormonal therapy
 - 5- Targeted therapy
 - 6- Immunotherapy
- Treatment depends on the *stage* of the disease and is tailored according the patient diagnosis:
 1. *Early stages*: Breast-conserving surgery (BCS) or modified radical mastectomy (MRM)
 2. *Late stages*: palliative MRM or primary chemotherapy followed by surgery
 - Oncoplastic BCS can be performed by one of the following ways:
 1. Volume displacement technique combines resection with glandular reshaping or reduction mammoplasty.
 2. Volume replacement technique combines resection with immediate reconstruction using an autologous tissue flap
 - Complications associated with axillary LN dissection:
 1. Neuropathy after surgery: the intercostal brachial (or "intercostobrachial") nerve is a branch of the second intercostal nerve.
 2. Motor injury: the thoracodorsal, long thoracic, pectoral nerves.
 3. Arm edema: it may cause substantial functional and psychological morbidity as well as predisposing patient to develop cellulitis and repeated infection of the upper arm.
 4. The more feared potential complication of arm edema is the development of lymphangio-sarcoma or "*Stewart-Treves syndrome*".

Sentinel Lymph Node Concept

- The concept of sentinel node is based on the fact that the efferent lymphatic channel draining a primary tumor will lead directly to the first sentinel-LN in the regional lymphatic basin. This lymphatic channel can carry malignant cells from the primary tumor to the sentinel LN. The tumor cells can then lodge in the subcapsular sinus in the LN and proliferate into a nodal metastasis. Thus, the sentinel node is the 'LN most likely to harbor metastatic disease if a regional nodal metastasis is present'.
- Approximately, 55% to 70% of patients undergoing sentinel LN biopsy may be spared possible complications of axillary dissection.
- Methods of sentinel lymph node mapping in breast cancer
 1. Blue dye technique
 2. Radio colloid technique
 3. Combined technique

Locally Advanced Breast Cancer (LABC)

- LABC is a heterogeneous group of tumors of varying clinical presentations and biological behavior. Some patients have a rapid neoplastic evolution, whereas others present with a long history of tumor growth.
- While in developing countries LABC represents up to 50% of the newly diagnosed breast cancers, in the USA it constitutes between 10% and 20% of all new breast cancers
- Locally advanced disease:
 - o T1 N2 - N3 M0
 - o T2 N2 - N3 M0
 - o T3 – T4 N1-N3 M0 (T3 > 7CM)
- Patients with LABC may be classified as stage IIB, III breast cancer according to the American Joint Committee (AJC) for Cancer.
- There is a great need for early treatment by systemic therapy (neoadjuvant chemotherapy aiming for down staging) for 3-4 cycles according to the response of the tumor. This is to be followed by surgery in responding patients and radiotherapy, followed by chemotherapy same regimen as before surgery.
- Also > 75% of these patients has clinically palpable axillary and/or supra-clavicular adenopathy, and 65%-90% of patients have pathologically confirmed LN metastasis; > 50% has > 4 LNs involved.
- *Clinical response to pre-operative chemotherapy*

Clinical response to pre-operative chemotherapy was defined on the basis of clinical and radiographic examinations as:

- 1- *Complete response* (total resolution of breast mass).
- 2- *Partial response* (> 50% reduction in the product of the two largest dimensions of the breast mass).
- 3- *Stable disease* (< 50% reduction in the breast mass).
- 4- *Progressive disease* (local: > 25% increases in tumor size or distant metastasis) during pre-operative chemotherapy.

Metastatic Disease

- Any T any N, M1
- Metastases can either be discovered at the diagnosis of the primary tumor or develop subsequently after free interval. The management of patients at this stage is determined by the duration of disease-free interval, the menopausal status, and the sites of metastases, and ER/PR status (chemo-hormonal therapy)
- Additional therapies may be needed in patients with metastatic disease, such as: fixation of pathological fracture, drainage of malignant pleural effusion, treatment of brain metastases by radiation and corticosteroids, and pain therapy by analgesics and narcotics.

Treatment of Recurrent Breast Cancer

1. Local recurrence after mastectomy:
 - Surgical excision and radiotherapy.
 - May consider systemic therapy.
2. Local recurrence after lumpectomy:
 - Mastectomy

Male Breast Cancer

- The classification of male breast cancer is the same as in females.
- Due to the lack of breast tissue, early invasion of the pectoralis fascia and muscle occurs. Hence, there is no place for conservation surgery in the male breast carcinoma.
- Mastectomy with axillary dissection is the primary mode of treatment in early cases.
- Adjuvant therapy depends on the stage of the disease and should follow the general guidelines for female patients.

Soft Tissue Neoplasms

The term soft tissue is used to describe all non-epithelial tissue that arise from any mesoderm derivative tissue except bone, cartilage, brain and its coverings, hematopoietic cells and lymphoid tissue.

CLASSIFICATION

Benign Tumors (more common)

- *Adipose tissue*: lipoma.
- *Fibrous tissue*: keloid, fibroma, nodular fasciitis
- *Fibrohistocytic*: fibrous histiocytoma
- *Muscle origin*: rhabdo-myoma, leiomyoma
- *Lymphatic system*: cystic hygroma, lymphangioma simplex, lymphangioma
- *Vascular system*: hemangioma, pyogenic granuloma, glomus body tumor
- *Nervous system*: neuroma, neurofibroma, Schwannoma
- *Uncertain (combined) origin*: hamartoma.

Locally Malignant (infiltrates locally, but do not metastasize)

- Desmoid tumor.
- Dermatofibrosarcoma protuberans

Malignant (fortunately rare)

- | | |
|----------------------------------|----------------------|
| - Liposarcoma | Fibrosarcoma |
| - Malignant fibrous histiocytoma | Rabdomyosarcoma |
| - Leiomyosarcoma | Lymphangiosarcoma |
| - Hemangiosarcoma | Malignant Schwannoma |
| - Kaposi sarcoma | Malignant hamartoma |
| - Synovial sarcoma | Epithelioid sarcoma |

Benign to Malignant Transformation

- Certain benign tumors are prone to undergo malignant changes, and it is important for both treatment and prognosis to realize when this occurs.
- Some or all of the following may be recognized:
 - 1- Rapid increase in size
 - 2- increased vascularity

- 3- Fixity
- 4- Involvement of nearby structure: e.g. facial palsy
- 5- Dissemination: to LNs or remote secondary's

BENIGN TUMORS

Lipoma

- It is the most common soft tissue neoplasm.
- It may occur anywhere in the body except where there is no adipose tissue (eye lid, glans penis). Head and neck, shoulder, abdominal wall and thighs are particularly favoured sites.
- It is usually solitary, but may present in families as multiple subcutaneous (SC) lipomatosis all over the body (*Dercum disease*).

Types

- *Pathological*: pure lipoma, fibrolipoma, hemangiolipoma, angiolipoma (painful).
- *Anatomical* (in order of frequency): subcutaneous, inter-muscular, retro-peritoneal, sub-fascial, sub-serous, sub-mucous, sub-synovial, intra-articular, para-osteal, CNS, intra-glandular.

Clinical Picture and Investigations

- *History*: A slowly growing, painless lump
- *Examination*: A lipoma is not tender. It is well-defined, with a smooth or lobulated surface and slippery edges, soft, freely mobile, pseudo-fluctuant and regional LNs are not enlarged.
- *Investigations* (diagnosis is usually on clinical base, but if needed)
 - 1- Fine-needle aspiration cytology (FNAC) → benign fat cells
 - 2- Ultrasound (US) → well-circumscribed, capsulated, solid swelling

Differential Diagnosis (DD)

- Sebaceous cyst
- Fibroma
- Organized hematoma
- Ganglion

Complications

- Sarcomatous changes (especially in certain sites: retro-peritoneal, back and thigh)
- Myxomatous degeneration
- Calcification
- Pressure on nearby structures
- Inflammation
- Disfigurement

Treatment

- Excisional biopsy is standard treatment.
- Liposuction could be helpful in cosmetic areas.
- In case of multiple tumors; excise the symptomatizing or complicating one.

Fibroma

- Pure fibroma is rare, most fibromas are combined such as fibromyoma, fibrolipoma, neurofibroma, etc
- It is either soft or hard depending on the proportion of fibrous to other cellular tissue.
- Some reactive fibroblastic proliferation may mimic benign tumor such as *nodular fasciitis* (eg. *Dipteran contracture*), which is usually a painless, small (< 3 cm), slowly growing mass, in the upper extremities and trunk.

Neurofibromatosis

- It is caused by autosomal dominant (AD) gene.
- There are two types:
 1. Type 1 (90%): It was previously known as *Von Recklinghausen disease*. characterized by multiple pedunculating nodules and pigmented skin discoloration (*café-au-lait spots*).
 2. Type 2: It was previously called *bilateral acoustic neurofibromatosis*.
- Clinically, there is pain, numbness, or pressure manifestation, tenderness and limitation of mobility along the axis of nerve where it originates from.

LOCALLY-MALIGNANT TUMORS

Desmoid Tumor

- It is an uncommon tumor that arises from musculo-aponeurotic structure.
- *Site*: The most common site is the anterior abdominal wall, which usually presents as a slowly growing, hard, mass in the anterior abdominal wall below the level of the umbilicus (more common in multipara women). Special intra-abdominal type is usually associated with familial polyposis coli (*Gardner syndrome*). It also occurs at the head and neck, thigh and shoulder.
- *Pathology*: formed of fibrous tissue with multinucleated plasmoidal masses, unencapsulated, infiltrate neighborhood muscle. Metastasis does *not* occur and no sarcomatous change happened.
- *Treatment*: wide excision with 1 inch safety margins all around. Muscular defect should be repaired with prosthetic mesh. Radiotherapy could be used as adjuvant or in cases where total excision is not possible.

Dermatofibrosarcoma Protuberans

- *Pathology*: It is an intermediate type between benign fibrohistioma and malignant fibrosarcoma. It is composed of a mixture of fibroblast and phagocytes (lipid laden cells with histiocytic appearance).
- *Clinical Picture*: It is a rare tumor, which occurs in young age usually in trunk and shoulders. It appears as slowly growing, bluish, dermal nodule. Suddenly it grows fast, ulcerates and infiltrates surrounding structures. Rarely, it gives metastasis to lungs.
- *Treatment*: Wide local excision in all direction including depth.

SOFT TISSUE SARCOMAS (STS)

- They are rare (1% of human cancer) and arise from mesenchymal tissue (mesoderm in origin).
- Although they have been traditionally classified according to tissue of origin, it is now evident that histological grade and clinical stage are more important as prognostic indicators and in determining the management approach.
- They differ from carcinomas by
 1. Younger age
 2. Rapid rate of growth
 3. Rapid local invasion
 4. Blood born metastasis is commoner
 5. Worse prognosis
 6. Less sensitive to radio & chemotherapy

Etiology

Unknown but there are some predisposing factors like:

1. Radiation exposure
2. Post mastectomy arm lymphedema
3. Von Recklinghausen disease

Grading

G1: Well differentiated

G2: Moderately differentiated

G3: Poorly differentiated

G4: Undifferentiated

Staging

T1: 5 cm or less

T2: > 5 cm

N0: No regional LN metastasis

N1: Regional LN metastasis

M0: No distant metastasis

M1: Distant metastasis

Differential Diagnosis

1. Benign soft tissue tumors
2. Organized hematoma
3. Vascular aneurysm and arterio-venous (AV) fistula
4. Lymphadenopathy
5. Myositis ossificans

Prognostic Factors (*the following make prognosis worse*)

1. *Type*: malignant histocytoma is worst
2. *Grading*: bad with undifferentiating
3. *Staging*: distant metastases are usually fatal
4. *Site*: retro-peritoneal is more fatal
5. *Local recurrence*: predisposes to further recurrence
6. *Biological expression* of tumor: e.g. p53

Clinical Picture

- Rapidly growing painless swelling, hot surface, ill-defined border, variable in consistency, fixed, and dilated subcutaneous veins.
- Enlarged LNs in special types.

Investigations

1. *CT*: Best in abdominal and thoracic lesions
2. *MRI*: Better in neurological, muscular and bony lesions
3. *PET*: Gives an idea about the metabolic function of sarcoma and surrounding tissue.
4. *Biopsy*: Either open incisional or better to be taken by FNAC or core tissue biopsy under US or CT guidance. (core biopsy is the preferred method)

Treatment

1. Surgical

- En-bloc wide excision in all directions should be performed, even if it includes major neurovascular structures, which need further reconstruction.
- Amputation is indicated in very huge, aggressive or recurrent cases.
- Excision of localized solitary metastases could be curative in some patients.
- LN block dissection is needed only in certain types of STSs (angiosarcoma synovial, Rhabdomyosarcoma, epithelioid)

2. Radiotherapy

- Preoperative: may ↓ size to facilitate surgical removal & ↓ local recurrence.
- Post-operative: as adjuvant in cases of suspected surgical margin involvement.

3. Chemotherapy

- Best results are achieved with childhood malignancy.

Characters of individual's Soft Tissue Tumors

Liposarcoma

- It is the commonest sarcoma in general, age of 50-60 years, most frequent in retro-peritoneal region and extremities.
- Types: well differentiated, myxoid, fibroblastic, lipoplastic, pleomorphic (arranged from better to worst prognosis).

Malignant Fibrohistosarcoma (Pleomorphic Undifferentiated Sarcoma)

- It is the most common sarcoma in young adult age, with 2nd peak between 50-70 years.
- It usually arises in the deep muscular tissue of extremities or in retroperitoneal space.
- It is a highly aggressive tumor that usually recurs locally and sends metastasis in > 50% of patients.

Fibrosarcoma

- *Incidence:* It is the 2nd most common sarcoma at the age of 35-50 years, higher incidence in the thigh and arm.
- *Clinically:* Firm, rubbery, painless nodule of long duration.
- *Histologically:* Low grade or high grade (worst prognosis).

Rhabdomyosarcoma

- It is predominantly a neoplasm of infancy and childhood.
- The most common histological variant is embryonal rhabdomyosarcoma with good response to variant therapeutic tools (others are alveolar and pleomorphic types).
- It arises usually in head and neck area, genitourinary tract and retro-peritoneum.

Leiomyosarcoma

- It is the most common sarcoma of the uterus and GIT.
- It is distinguished from leiomyoma by the infiltrate growth, great cellularity, pleomorphism, greater mitotic activity and the presence of metastasis.

Kaposi Sarcoma

- It is a rare malignancy characterized by flat purplish-red skin lesions particularly lower limb that may affect also mucous membrane, GIT and other visceral organs.
- There is well known two types:
 1. Non-aggressive form, affecting mainly Central European Jews.
 2. Aggressive form, affecting mainly immuno-suppressant patients (AIDS, organ transplantation). Characterized by fever, chills, internal organ involvement and lymphadenopathy.
- Treatment: AZT and interferon (IFN) to raise the immunity of AIDS patients, radiotherapy, and surgery (limited to small localized lesion and for treatment of intestinal complications).

Synovial Sarcoma

- This tumor accounts for 10% of all STSs, known also as *spindle cell tumor*.
- Despite of the name, it does *not* arise from synovial cells; rather, it derives from mesenchymal cells about joint cavities & sometimes in sites totally remote from joint.
- It has a variable histological grading but the high grade one is an aggressive tumor with high recurrence rate (RR).

RETR-PERITONEAL TUMORS

Surgical Anatomy

- The space between the posterior abdominal muscular wall and peritoneum is termed *retoperitoneal space*, which extends from the diaphragm above to the pelvic brim below.
- It contains: Aorta, IVC, cysterna chili, para-aortic LNs and vessels, lumbar plexus and chain of sympathetic ganglia. The kidneys, ureters, adrenals and pancreas are also within this space. The term retro peritoneal tumor is usually confined to tumors arises in retro peritoneal space with *exclusion* of kidneys, adrenals and pancreatic swellings.

Classification and Examples

- *Cystic*: Teratoma, remnant of Wolffian ducts, dermoid, parasitic, lymphogenous.
- *Solid benign* (20%): lipoma, leiomyoma, neurofibroma.
- *Malignant* (80%): lymphoma, liposarcoma, fibrosarcoma, neurofibrosarcoma.

Diagnosis

- *History*: *Vague symptoms*: abdominal discomfort, back pain, anorexia, fatigue and weight loss.
- *Examination*: *Abdominal mass*: In most cases, a *mass* could be palpated. In some huge tumors obstruction of IVC leads to edematous legs, dilated abdominal veins and ascites.
- *Imaging*: CT and MRI are the investigations of choice.
- *CT-guided core biopsy*: has to be done in all cases

Treatment

- *Surgical excision* is the gold standard treatment. However, only 25% of tumors could be completely resected.
- In irresectable cases: Excision of as much as possible from the tumor (*debulking*) combined with *radio/chemotherapy* carries a better prognosis than either alone.

Retro-Peritoneal Tumors

SURGICAL ANATOMY

- The retro-peritoneum represents a complex potential space with multiple vital structures bounded anteriorly by the peritoneum, ipsilateral colon and mesocolon, pancreas, liver or stomach.
- The posterior margins are by large composed of the psoas, quadratus lumborum, transverse abdominal and iliacus muscles but, depending on the tumor location and size, may be formed by the diaphragm, ipsilateral kidney, ureter and gonadal vessels.
- Similarly, the medial boundaries may include the spine, para-spinous muscles, the inferior vena cava (for right-sided tumors) and the aorta (for left sided tumors).
- The lateral margin is formed by the lateral abdominal musculature and, depending on tumor location, may include the kidney and colon.
- Superiorly, retro-peritoneal tumors may be in contact with the diaphragm, the right lobe of the liver, the duodenum, the pancreas or the spleen.
- The inferior margin may relate to the iliopsoas muscle, the femoral nerve, the iliac vessels or pelvic sidewall
- The retro-peritoneum can host a wide spectrum of pathologies, including a variety of rare benign tumors and malignant neoplasms that can be either primary or metastatic lesions.
- Malignant tumors of the retro-peritoneum occur four times more frequently than benign lesions.
- Sarcomas comprise a third of retroperitoneal tumors.

RETO-PERITONEAL SARCOMA

Pathology

- Sarcomas constitute a heterogeneous group of rare solid tumors of mesenchymal cell origin with distinct clinical and pathological features.
- They are divided into two broad categories; soft tissue sarcomas (STS) and bone sarcomas
- Sarcomas collectively account for approximately 1% of all adult malignancies and 15% of pediatric malignancies.

- The true incidence of STS is underestimated, especially because a large proportion of patients with gastrointestinal stromal tumors (GISTs) may not have been included in tumor registry database before 2001.
- The most common subtypes of STS are undifferentiated pleomorphic sarcoma, GISTs, liposarcoma, leiomyosarcoma, synovial sarcoma and malignant peripheral nerve sheath tumors.
- The anatomical site of the primary disease represents an important variable that influences treatment and outcome. Extremities (43%), trunk (10%), visceral (19%), retroperitoneal (15%) and head and neck (9%) are the most common primary sites. Approximately 80% of the neoplasms that arise within the retroperitoneal space are malignant.
- Furthermore, the majority of patients who present with a primary retro-peritoneal, extra-visceral, uni-focal soft tissue mass will be found to have a sarcoma. Retroperitoneal sarcoma represents only a third of retroperitoneal tumors.
- In adults, the most common histologic types of retroperitoneal STS are lipo-sarcomas (70%) and leiomyo-sarcomas (15%), followed by pleomorphic undifferentiated sarcoma/malignant fibrous histiocytoma.
- A variety of other histologic types may be observed, but they are much less common in the retro-peritoneum than in other primary sites.

Liposarcoma

- Types
 1. Well differentiated liposarcoma
 2. Dedifferentiated liposarcoma
 3. Pleomorphic liposarcoma

Leiomyosarcoma

- Leiomyosarcoma (LMS) is sarcoma showing smooth muscle differentiation; this is based on a combination of histomorphology and immuno-phenotypic expression.
- In contrast to other sarcomas, the use of core-needle biopsy specimens in the primary diagnosis of LMS is increasingly recognized to correlate poorly with final tumor grade in resection specimens and this may have important implications in treatment

Other Types of Sarcoma

1. Malignant peripheral nerve sheath tumors
 2. Synovial sarcoma
 3. Solitary fibrous tumor (formerly known as a hemangio-pericytoma)
 4. Ewing and Ewing-like sarcoma
 5. Desmoplastic small round cell tumor
 6. Undifferentiated/Unclassified sarcoma
- Among children, the most common histological types of retro-peritoneal STS are extra-skeletal Ewing sarcoma/primitive neuroectodermal tumors [PNET], alveolar rhabdomyosarcoma, and fibrosarcoma

- Approximately one-half of all retro-peritoneal sarcomas are high-grade tumors, although this varies according to histology. The majority of retro-peritoneal liposarcomas are low- to intermediate-grade lesions.
- STS most commonly metastasize to the lungs; tumors arising in the abdominal cavity more commonly metastasize to the liver and peritoneum.
- Management of STS in adult patients is addressed from the perspective of the following disease subtypes:
 1. Retro-peritoneal or intra-abdominal STS.
 2. GISTs
 3. Desmoid tumors
 4. Rhabdomyosarcoma.

Genetic Cancer Syndromes with Predisposition to STS

- Li-Faumeni syndrome (TP53 germline mutation). Associated with RMS, fibrosarcoma and undifferentiated pleomorphic sarcoma.
- Gardner syndrome (desmoid tumors)
- Carney-Stratakis syndrome (gastric stromal sarcoma and paragangliomas) and Carney's triad (gastric stromal sarcoma, paraganglioma and pulmonary chondroma)

Work-up

- Prior to the initiation of therapy, all patients should be evaluated and managed by multidisciplinary team (MDT).
- History and physical examination.
- Chest, abdomen and pelvis CT with contrast (Figure 1)
- MRI may add some data

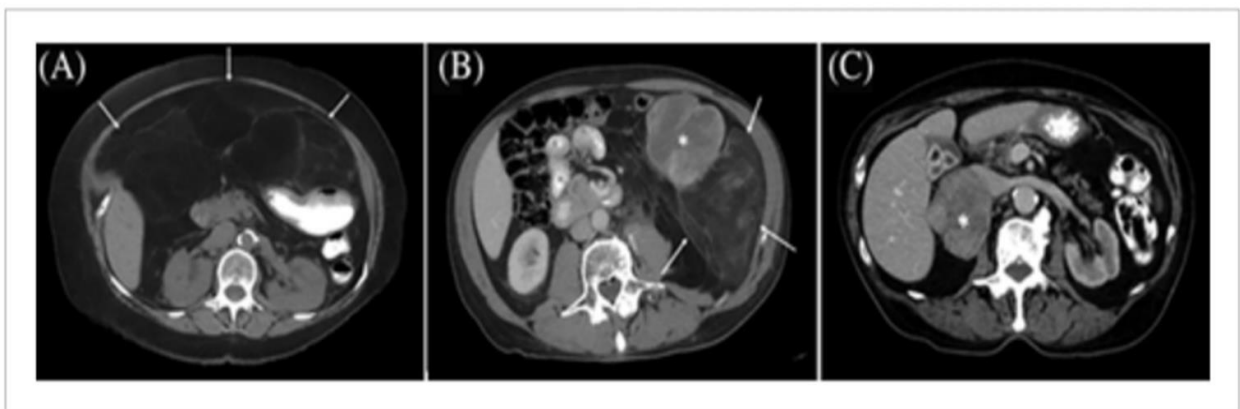


Figure 1: CT axial image of a homogeneous fat dense mass consistent with well-differentiated liposarcoma (arrows). Axial image of a dedifferentiated liposarcoma with a solid mass (asterisk) with surrounding fat dense well-differentiated component (arrows).

Criteria for Unresectability

Radiographic findings that indicate unresectability include:

- Extensive vascular involvement (aorta, vena cava and/or iliac vessels), although involvement of the vena cava and iliac veins is a relative rather than absolute contraindication, as these vessels can often be replaced with interposition grafts
- Peritoneal implants
- Distant metastases
- Involvement of the root of the mesentery (specifically, the superior mesenteric vessels)
- Spinal cord involvement
- Biopsy:
 - Pre-resection biopsy is not necessarily required ; consider biopsy if there is suspicion of malignancy other than sarcoma.
 - Image-guided (U/s or CT) core needle biopsy is preferred
- Patients with personal/family history suggestive of Li-Fraumeni syndrome should be considered for further genetic assessment.

Staging

- Retro-peritoneal sarcomas are staged using the same TNM system as is used for extremity STS. However, the ability of the TNM staging system to discriminate outcomes is limited.
- Several studies have found no prognostic role for tumor size in retroperitoneal sarcoma.
- Given the importance of histologic grade and resection margins in the prognosis of retroperitoneal STS, an alternative staging system has been proposed that incorporates these features as well as the presence or absence of metastatic disease. However, this staging system is not in widespread use.

Primary tumor (T)

TX	Primary tumor cannot be assessed
T0	No evidence for primary tumor
T1	Tumor $\leq$ 5 cm in greatest dimension
T2	Tumor > 5 cm to $\leq$ 10 cm in greatest dimension
T3	Tumor > 10 cm to $\leq$ 15 cm in greatest dimension
T4	Tumor is > 15 cm

Regional lymph nodes (N)

N0	No regional lymph node metastasis or nodes cannot be assessed
N1	Regional lymph node metastasis

Distant metastasis (M)

M0	No distant metastasis
M1	Distant metastasis

Histologic grade (G)

GX	Grade cannot be assessed
G1	Grade 1
G2	Grade 2
G3	Grade 3

Stage IA	T1	N0	M0	G1, GX
Stage IB	T2	N0	M0	G1, GX
	T3	N0	M0	G1, GX
	T4	N0	M0	G1, GX
Stage II	T1	N0	M0	G2, G3
Stage IIIA	T2	N0	M0	G2, G3
Stage IIIB	T3	N0	M0	G2, G3
	T4	N0	M0	G2, G3
	Any T	N1	M0	Any G
Stage IV	Any T	Any N	M1	Any G

Treatment

Surgical Resection

- Surgical resection has traditionally been the only potentially curative treatment for a localized retroperitoneal STS.
- The ability to perform a complete surgical resection at the time of initial presentation is the most important prognostic factor for survival.
- The primary oncologic goal is complete resection with microscopically negative margins (R0 resection).
- However, the large size of most retroperitoneal tumors, coupled with the inability to obtain wide margins due to anatomic constraints make this goal difficult to achieve. In clinical practice, many resections are grossly complete but with microscopically positive margins (R1 resection)
- Resection of adjacent organs such as the small bowel, colon or kidney is often required to achieve a complete resection and bowel preparation and evaluation of kidney function should be performed prior to exploration.
- Liberal en-bloc resection of adjacent viscera may allow a subset of patients to achieve wide, macroscopically negative surgical margins who might otherwise have been considered unresectable.
- *There is no survival benefit for incomplete resection (a "debulking" procedure) in patients with unresectable retroperitoneal STS.*

Adjunctive Radiotherapy (RT)

- In contrast to extremity STS in which the most common site of first recurrence is a distant site, the primary pattern of treatment failure after resection of a retroperitoneal STS is local.
- Adjunctive radiation therapy (RT) can be administered following resection (adjuvant RT). However, increasingly, preoperative RT is being chosen for large high-grade or intermediate-grade STS.

Adjuvant RT

- It is often difficult to deliver postoperative radiation therapy because the bowel and other organs fall into the resection cavity; however, newer techniques such as intensity-modulated RT (IMRT) and proton beam irradiation make it more feasible, but the therapeutic ratio is probably still more favorable with preoperative RT. Nevertheless, it is reasonable to consider the use of postoperative RT if it can be delivered safely.
- In the post-operative setting, radiation doses to the tumor bed are often limited by the large field size and the proximity and tolerance of surrounding radiosensitive normal structures, such as the liver and bowel.
- In fact, many multidisciplinary sarcoma groups do not routinely offer postoperative RT to patients with resected retroperitoneal sarcomas because of significant concerns about the narrow therapeutic ratio.

Pre-operative RT

- The delivery of RT prior to surgery, with or without intraoperative RT (IORT) at the time of resection, may permit the safe delivery of higher RT doses than are possible in the postoperative setting

Outcomes and Prognostic Factors

- *Retroperitoneal sarcomas have a substantially less satisfactory outcome than soft tissue sarcomas (STS) at other sites, such as the extremities or trunk.*
- Several factors contribute to poor outcome and a high rate of recurrence:
- The surrounding normal tissues (liver, kidney, gastrointestinal tract, spinal cord) have relatively low tolerance for radiation therapy (RT). As a result, radiation dose levels must be kept below those typically employed for extremity sarcomas.
- In contrast to extremity sarcomas, 90% of first recurrences are local. Eventually, distant metastases develop in 20-30%. The main sites of distant metastases are liver and lungs.
- Local recurrence rates are higher with high-grade (poorly differentiated) tumors, liposarcoma histology, and in patients with positive resection margins.

CHAPTER

5

Lymphadenopathy and Lymphoma

SURGICAL ANATOMY

Lymphatic System

- Lymphatics are present in all organs of the body except brain, eyes, bone marrow and cartilage
- Thin-walled lymphatic vessels progressively increase in caliber
- About 600 lymph nodes (LNs) are present in the body
- Slow flow, low pressure system

Function

- Interstitial fluid balance
- Filters antigens from the interstitial fluid
- Production of lymphocytes and macrophages
- Primary site of immune response to tissue antigens
- Maturation of lymphocytes

Lymph Nodes (Figure 1)

- They are small encapsulated organs
- They lie along the lymph vessels
- They serve as filters for lymph on its way to the blood
- Antigen-activated lymphocytes differentiate in lymph nodes

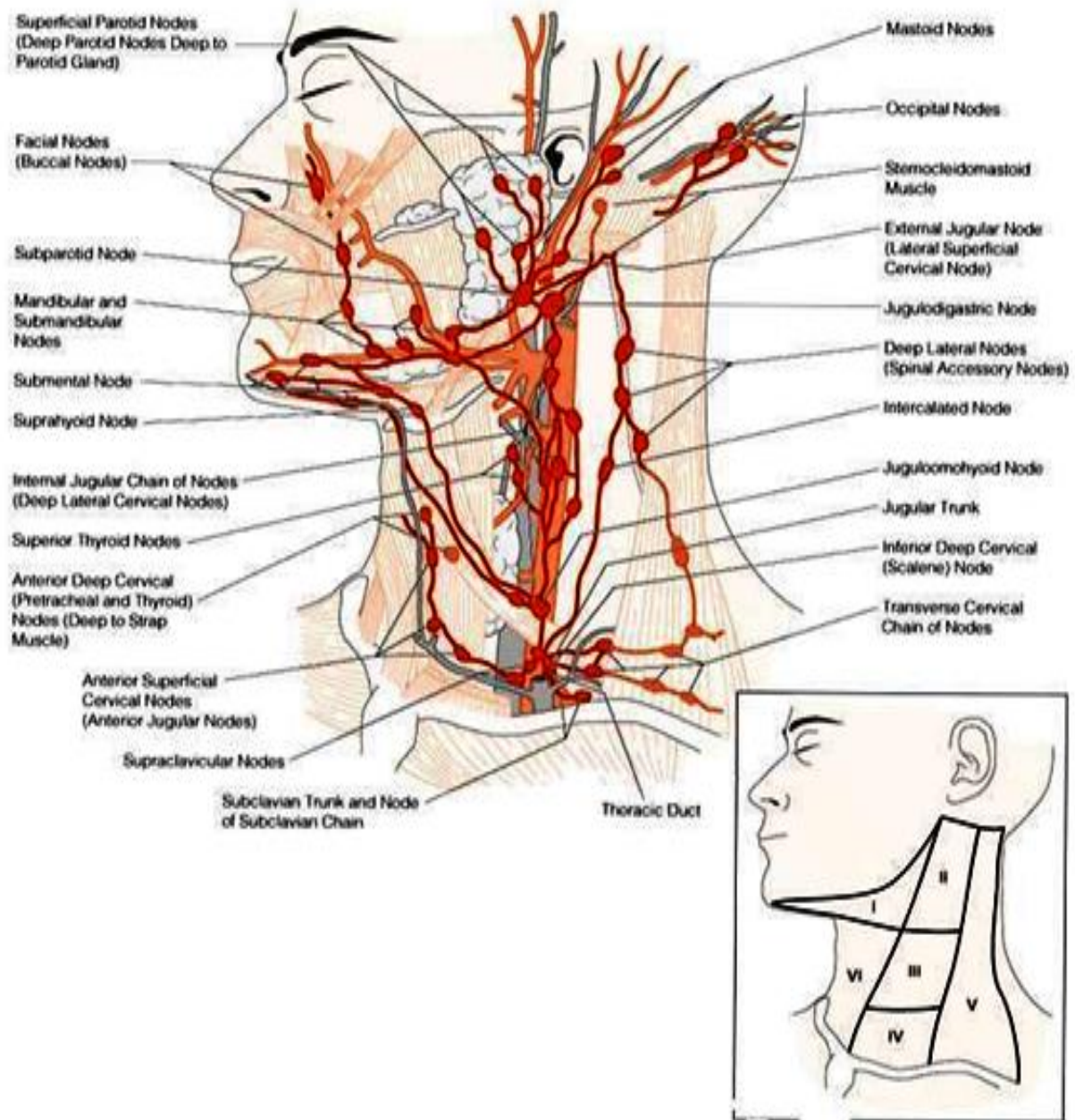


Figure 1: Distribution of cervical lymph nodes (levels I-VI)

LYMPHADENOPATHY

Definition

- Lymphadenopathy means ‘lymph nodes that are abnormal in size, consistency or number’. It may be localized (75%) or generalized (25%)

Causes of Localized Lymphadenopathy

1. *Inflammation*: Acute - chronic specific - chronic non-specific
 2. *Neoplasms*: Early lymphoma - local metastasis
- Most cases are benign and self-limited
 - The *challenge* is to decide if it is representative of a serious illness or not (LN biopsy)
 - Distinguish between benign and malignancy according to:
 1. Age
 - o Age <30 years: 79% benign, 15% lymphomas, 6% carcinomas
 - o Age >50 years: 40% benign, 16% lymphomatous, 44% carcinomas
 - o Age 30-50 years: indeterminate value
 2. Character
 3. Location

Lymph Node Characters

- Pain and tenderness, rapid increase in size, stretch of capsule indicate inflammation, suppuration, or hemorrhage
- Size does not indicate a specific disease process
- Consistency:
 - o Stone hard: cancer usually metastatic
 - o Firm rubbery: lymphoma
 - o Soft: infection or inflammation
- Shotty: Non-specific
- Fluctuating nodes: suppuration
- Matting: Tuberculosis (TB)

Famous Lymph Nodes

- *Jugulo-digasric*: Tonsillitis, pharyngitis
- *Virchows*: Left supra-clavicular (abdominal or thoracic cancer)
- *Delphian*: Pre-laryngeal (thyroid or laryngeal cancer)
- *Node of Cloquet (Rosenmuller node)*: Deep inguinal near femoral canal

Causes of Lymphadenopathy

Cancer

- *Heme malignancies*: Hodgkin disease, non-Hodgkin lymphoma (NHL), acute and chronic leukemias, multiple myeloma
- *Metastatic*: solid tumor (breast, lung, renal, ovarian)

Hypersensitivity Syndromes

- Serum sickness
- Serum sickness-like illness
- Silicone
- Vaccination
- Graft versus Host disease (GVHD)
- Drugs: Cephalosporins, Atenolol, Captopril, Sulfonamides, Allupurinol

Infections

Viral

- Epstein-Barr virus (EBV): Mono spot test
- Cytomegalovirus (CMV): immunosuppressed, transplant recipient, recent blood transfusion
- Human immuno-deficiency virus (HIV): IV drug abuse, high-risk sexual behavior
- Hepatitis: IV drug use
- Herpes Zoster: Superficial cutaneous nodules

Bacterial

- Staph/strep: cutaneous source, lymphadenitis
- Cat scratch disease
- Mycobacterium: TB and non-TB

Protozoan

Mycotic

Rickettsial (typhus)

Helminthic (filariasis)

Connective Tissue Disease

- Rheumatoid arthritis (RA)
- Systemic lupus erythematosus (SLE)
- Dermatomyositis
- Mixed connective tissue disease
- Sjogren's disease

Atypical Lymphoproliferative Disorders

- Castleman's disease
- Angio immunoblastic lymphadenopathy with dys-proteinemia

Granulomatous Diseases

- Histoplasmosis
- Mycobacterial infections
- Cryptococcus
- Cat Scratch disease

Others

- Kawasaki disease

Algorithm to Evaluate Lymphadenopathy

- Attention to history and physical exam
- Confirmatory testing
- Indication for biopsy

History

- Localizing symptoms or signs to suggest a specific site
- Constitutional symptoms: B-lymphoma symptoms: fever, night sweats, >10% body weight loss in < 6months)
- Epidemiological clues: occupation, travel, high risk behavior
- Medications

Non-specific Chronic Lymphadenopathy

- Children Girls, Cervical, Fluctuant in size, Painless, Firm and Shotty
- Treatment: Assurance - Wait 3-4 weeks and re-examine

Non-specific Acute Infection

- No indication for empiric antibiotics or steroids
- Steroids can be harmful and delay diagnosis

Metastatic LN

- Hard
- Irregular
- Fixed
- Evidence of primary

Metastatic LN with Occult Primary

- *Head and neck:* Larynx and pharynx, posterior third of tongue, thyroid gland (papillary type), airway sinuses.
- *Chest:* Bronchial carcinoma, esophagus.
- *Abdomen:* GIT (stomach, pancreas and colon).
- *Urinary:* Kidney and prostate.
- *Genital:* Testes and ovaries.
- *Limbs:* Melanoma.

Biopsy

- It can be done by surgery, mediastinocopy or laparoscopy
- Fine-needle aspiration (FNA) is *not* recommended as it cannot distinguish between lymphomas (nodal architecture needs to be intact)
- FNA is reserved for established diagnosis and to demonstrate recurrence
- Ideally, axillary and inguinal nodes are avoided as often demonstrate reactive hyperplasia
- Preferred: supra-clavicular, cervical, axillary, epitrochlear, inguinal
- Complications include vascular and nerve injury

LYMPHOMA

Definition

- Lymphoma is a primary malignant disease of the lymphoid system

Role of Surgery in Lymphoma

- Lymph node biopsy.
- Excision of residual LNs after radiotherapy or chemotherapy.
- Resection of visceral lymphoma (gastric)
- Staging laparotomy (obsolete).
- Management of complications:
 1. Bleeding.
 2. Obstruction.
 3. Perforation
 4. Compression fracture.

Etiology and Risk Factors

Risk factors associated with this disease:

- *Age*: Hodgkin's disease occurs with a main peak at 15-35 years. Rare at > 55 years.
- *Gender*: More common in men.
- *Family History*: Brothers and sisters of those with Hodgkin's disease have ↑ incidence (7 folds).
- *Viruses*: Epstein-Barr virus infection ↑ chance of getting Hodgkin's disease.

Hodgkin Lymphoma (HL)

Pathology

Histopathologically, the lymph nodes in Hodgkin's disease are characterized by:

- Reed-Sternberg (R-S) cells: Dominant in all sub-types. Large cells with abundant basophilic cytoplasm. Bi-nucleated or multinucleated (mirror image appearance).
- Reactive lymphocytes.
- Benign histiocytes, plasma cells and eosinophils.

Presentation

Adenopathy

- A painless swelling (LNs in the neck, axilla, or groin).
- Common site is upper deep cervical LN.
- Characters: Painless, discrete, rubbery in consistency

Systemic Manifestations

- Develops in 25-35% of patients
- Unexplained weight loss of > 10% of body weight in the 6 months before diagnosis.
- Skin itching and pruritis ($\pm$)
- Unexplained fever (Pel-Ebstein fever): Unexplained intermittent with temperatures above 38°C , alternating with afebrile period, and lasting for days or weeks.
- Drenching night sweats

Cellular Classification

Revised European-American Lymphoma (REAL) classification is still the histological classification of Hodgkin lymphoma (HL)

- Proportion of R-S cells to lymphocytes.
 1. Lymphocyte-rich classical HL.
 2. Mixed cellularity HL.
 3. Nodular sclerosis HL.
 4. Lymphocyte depleted HL.
- Increase proportion R-S cells means a more aggressive HL.

Non-Hodgkin Lymphoma (NHL)

Definition

- It is a heterogeneous group of lymphoproliferative malignancies characterized by expansion of malignant B and T lymphocytes, with differing patterns of behavior and responses to treatment, and greater predilection to disseminate to extra-nodal sites.

Presentation

Lymphadenopathy (2/3 of patients)

- Painless, huge, tendency to amalgamation.
- Signs of hyper-vascularity: hotness, dilated veins.
- Location:
 - Head and neck: Waldeyer's ring, upper deep cervical LNs.
 - Chest: cough, pain SVC compression.
 - Abdomen: asymptomatic, abdominal mass, splenomegaly.
 - GIT manifestations as pressure manifestations, surgical emergencies.

Extra-nodal Involvement

- Arises in tissues or organs separate from, but near, major lymphatic aggregates.
- 10-25% first presentation and 50% along the course of the disease.
- Common sites: GIT, testes, kidney, CNS, skin and bone.
- Testes: most common malignancy after the age of 60 years.
- Bone: Pain, swelling, fracture.
- CNS: one of the most primary CNS tumours. Cord compression (5%).

Systemic Manifestations (B-symptoms)

- 50% with aggressive lymphoma
- 25% with indolent lymphoma
- Its presence indicates bulky disease.
- Fatigue and malaise, not B-symptoms (10%).

Pathology – Cell Types

- Follicular and diffuse small cleaves and large cell.
- Immunoblastic.
- Lymphoblastic (LL).
- Small non-cleaved cell

Burkitt's Lymphoma

- It involves younger patients, the most common type of pediatric NHL.
- It is aggressive extra-nodal B-cell lymphoma.
- It has a 20% to 30% life-time risk of CNS involvement
- *Endemic cases*: Usually in Africa. Involve the facial bones or jaw of children.
- *Sporadic cases*: Usually involve the gastrointestinal system, ovaries, or kidneys, and present with mass.
- *Treatment*: It involves aggressive regimens as the advanced-stage aggressive NHL. Surgical resection of complicated or symptomatic mass.

Mycosis Fungoides (Sezary Syndrome)

- Classified as cutaneous low grade T-cell lymphoma.
- It is an indolent condition. Symptoms present for long periods (average of 2-10 years) of cutaneous eruptions.
- Cutaneous disease typically progresses from an eczematous patch/plaque stage:
 - o (T1): plaque stage covering < 10% of body surface
 - o (T2): plaque stage covering > 10% of BS.
 - o (T3): tumors frequently undergo necrotic ulceration.
 - o (T4): SS is an advanced form of MF with generalized erythroderma and peripheral blood involvement.
- Skin biopsy: Presence of MF/SS cells (convoluted lymphocytes) + epidermal infiltrations with of neoplastic lymphocytes.

